Ruiquan (Richard) Su

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Education

University of Waterloo — Waterloo, ON
Computer Engineering, Hon. Coop, President's Scholarship with Distinction, GPA: 4.0/4.0 — *Sept. 2024 - Present*

Experience

Fullstack Engineer — June 2026 - August 2026
RELEX Solutions — *Toronto, ON*

- Designed and built the enterprise **Knowledge Base platform** using Flask, React, PostgreSQL, Socket.io, and AWS to support **AI business data synthesis**, enabling configurable **database-backed context**, schemas, business rules, memories, RBAC, real-time collaboration, and data validation.
- Developed KB **Memories**, an example-retrieval layer that curates successful chat outcomes into **memory artifacts**, with **similarity detection** and automatic **revalidation** as the KB evolves.
- Implemented KB lifecycle and version-control workflows alongside various AI-assisted KB authoring pipelines.

Backend Developer — September 2025 - December 2025
Lynkr Inc. — *Toronto, ON*

- **Co-developed**, with one other developer, the company's **core** product, Lynkr Workbench, from **scratch** to **beta**, generating over **500,000 CAD** in SaaS revenue and serving **2,800+** individual customers.
- Designed the backend **architecture** with **FastAPI** and **Next.js**, and built the **Enterprise tier** featuring **RBAC**, member management, SAML SSO, seat control, Stripe billing, etc.
- Built dual **LangGraph** LLM pipelines supporting autonomous AI **agents**, MCP service calls, LangFuse billing tracker, and categorized long-term **memory**.

Control Software Developer — January 2025 - April 2025
SIOUX Technologies (Asia), Jiangsu Industrial Technology Research Institute (JITRI) — *Suzhou, China*

- Cut on-site deployment time **from 98 hours to 20 hours** by designing and developing concurrent **streamlined installation and upgrade processes**, reducing manual overhead via GitLab **CICD pipelines** and **scripting**.
- Refactored existing services into **organized packages** and developed **Blazor-based API controllers** to automate installations, enabling **direct customer access**.
- Organized and led **company-wide Git training sessions** to enhance version control proficiency across teams.

Project Selection

Reimburst | *Next/TypeScript, FastAPI, PostgreSQL, Redis, Azure* — June 2026 - Present

- Built a full-stack **reimbursement automation service** that uses **computer vision/OCR** to extract receipt merchants, dates, totals, taxes, and line items, then categorizes expenses using an **LLM**.
- Developed an **asynchronous processing pipeline** with **FastAPI, Redis, and background workers**, supporting receipt extraction, expense categorization, and synopsis generation.
- Implemented **role-based access control** and a reimbursement **state machine** for employees, approvers, and administrators, with **immutable audit logs, reconciliation checks, and idempotent payout processing** to maintain financial integrity.

SnapLearn | *Flask, PyPDF, SciKit-learn, Next.js, Auth0, spaCy* — September 2025 - October 2025

- Built a **Next.js and Flask** application that converts uploaded textbooks and users' study schedules into personalized learning curricula.
- Designed a Python **NLP algorithm** using **cosine similarity** to model prerequisite relationships between topics as a directed acyclic graph.
- Developed a **greedy scheduling algorithm** to allocate topics and practice sessions within user schedules, and trained a **small language model** to estimate study time.

Technical Skills

Languages: Python, C/C++, C#, SQL, JavaScript/TypeScript, PowerShell, YAML
Frameworks & Libraries: FastAPI, Flask, Next.js, React, Pydantic, NumPy, pandas, spaCy, TensorFlow, Celery
Cloud & Infrastructure: AWS, Azure, Terraform, Docker, PostgreSQL, DynamoDB, Redis
Developer Tools: Git, GitHub Actions, Node.js, VS Code, Jira, Jupyter